

Adaptive Hedge

Adapting the learning rate to the difficulty of the data

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Statistical Methods for Machine Learning — Experimental Projects

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A.Y. 2025/26

Paper: T. van Erven, P. Grünwald, W. M. Koolen, S. de Rooij, *Adaptive Hedge*, NeurIPS 2011
([arXiv:1110.6416](https://arxiv.org/abs/1110.6416))

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What is in this report

The Hedge algorithm has one parameter, the learning rate η , and its performance depends on it entirely. The standard tunings pick η to protect against the worst possible loss sequence, which makes them far too cautious whenever the data happen to be easy. AdaHedge removes the parameter: it measures, online, how hard the data actually are, and lowers η only when the measurement says it must. This report reimplements Hedge and AdaHedge from scratch and asks whether that measurement does what the paper claims, on six loss sequences of controlled difficulty.

Route.

1. **The problem** (§1) — protocol, regret, and why the learning rate is the whole difficulty.
2. **The algorithms** (§2) — mixability gap, budget rule, AdaHedge, and the six results from the paper that the experiments test.
3. **Implementation** (§3) — what was written from scratch, the one numerical subtlety that matters, three correctness checks.
4. **Design** (§4) — six regimes, 50 seeds, and what is averaged over what.
5. **The AP construction** (§5) — sequences with a leader switching on a known schedule, in three variants separating the *size* of the signal from its *growth rate*.
6. **Results** (§6) — the regret table and the required plots.
7. **Theory against measurement** (§7) — the chain posterior $\rightarrow$ mixability gap $\rightarrow$ segments $\rightarrow$ regret, with every link measured.
8. **Conclusions** (§8) — including where AdaHedge loses, and why.

1 The problem, and why the learning rate is the difficulty

Definition 1 (Decision-theoretic online learning). A set of K actions is fixed. At each round $t = 1, \dots, T$ the learner publishes a probability vector $\mathbf{w}_t$ over the actions; the environment then reveals a loss vector $\ell_t \in [0, 1]^K$, and the learner pays $\mathbf{w}_t \cdot \ell_t$. After T rounds the learner's *regret* is

$$R(T) = \sum_{t=1}^T \mathbf{w}_t \cdot \ell_t - L_T^*, \quad L_T^* = \min_{1 \leq k \leq K} L_T^k, \quad L_T^k = \sum_{t=1}^T \ell_t^k.$$

Note — *Three things this report does not call by the same name.* Three distinct actions recur below and must be kept apart — conflating the first two is the easiest available mistake here. (i) The **best action** is $\arg \min_k L_T^k$, fixed over the whole horizon: the comparator inside $R(T)$, and what *rank 1* means in the weight plots. (ii) The **current leader** is $\arg \min_k L_t^k$, recomputed each round: the comparator in the regret-*vs.*-*time* curves, and the action carrying w_t^* . It flickers. (iii) The **window leader** (§5 only) is the action whose losses AP thins inside a given window. It is set by the construction, not by the data.

Hedge plays $w_{t+1}^k \propto e^{-\eta L_t^k}$. Writing $B_t = \sum_k \frac{1}{K} e^{-\eta L_t^k}$, the weights are a Bayesian posterior with uniform prior and pseudo-likelihood $e^{-\eta L_t^k}$, and B_t plays the role of the marginal likelihood. The regret of Hedge decomposes exactly into two terms, and that decomposition is the object the whole paper manipulates:

$$R(T) \leq \frac{\ln K}{\eta} + \Delta_T(\eta), \quad \Delta_T(\eta) = \sum_{t=1}^T \delta_t(\eta), \quad \delta_t(\eta) = \mathbf{w}_t \cdot \boldsymbol{\ell}_t + \frac{1}{\eta} \ln(\mathbf{w}_t \cdot e^{-\eta \boldsymbol{\ell}_t}). \quad (1)$$

The two terms pull in opposite directions. Raising η shrinks $\ln K/\eta$, the price of not knowing which action is best; it inflates $\Delta_T(\eta)$, the price of committing too fast. The classical tunings $\eta = \sqrt{2 \ln K/T}$ or $\eta = \sqrt{2 \ln K/L_T^*}$ (eq. (9) of the paper) balance the two *for the worst case*. On data where the best action is obvious, the second term never grows, the balance is struck at the wrong point, and the learner pays $\ln K/\eta$ for caution it did not need. That is the gap AdaHedge is designed to close.

Definition 2 (Mixability gap). $\delta_t(\eta)$ is the amount by which Hedge's actual loss exceeds $-\frac{1}{\eta} \ln(\mathbf{w}_t \cdot e^{-\eta \boldsymbol{\ell}_t})$, the loss of the Aggregating Pseudo-Algorithm, which mixes the actions with ideal efficiency and is not realisable. It is a per-round measure of how far the data force Hedge from that ideal, and it is observable at run time — which is what makes it usable as a tuning signal.

2 The two algorithms and the results being tested

Lemma 3 (Paper's Lemma 1). *For every t and every $\eta > 0$, $0 \leq \delta_t(\eta) \leq \eta/8$.*

Lemma 4 (Paper's Lemma 2). *For every T and $\eta \in (0, 1]$, $\Delta_T(\eta) \leq (\eta L_T^* + \ln K)/(e - 1)$.*

Lemma 4 says that when $\Delta_T(\eta)$ becomes large the learning rate must already be at least $\sqrt{(e - 1) \ln K/L_T^*}$. AdaHedge turns this around: rather than solving for η from an unknown L_T^* , it fixes a *budget*

$$b(\eta) = \left(\frac{1}{e-1} + \frac{1}{\eta} \right) \ln K \quad (2)$$

and runs at rate η until $\Delta_t(\eta)$ exhausts it, at which point it restarts with $\eta \leftarrow \eta/\phi$. Restarting also resets the weights to uniform, because $\mathbf{w} \propto e^{-\eta \mathbf{L}}$ encodes the old η and cannot be carried across.

Algorithm 1 AdaHedge(ϕ), $\phi = 2$

```

 $\eta \leftarrow \phi$ 
for  $t = 1, 2, \dots$  do
  if  $t = 1$  or  $\Delta \geq b$  then ▷ start a new segment
     $\eta \leftarrow \eta/\phi$ ;  $b \leftarrow \left( \frac{1}{e-1} + \frac{1}{\eta} \right) \ln K$ ;  $\Delta \leftarrow 0$ ;  $\mathbf{w} \leftarrow \left( \frac{1}{K}, \dots, \frac{1}{K} \right)$ 
  end if
  output  $\mathbf{w}$ ; observe  $\boldsymbol{\ell}_t$ 
   $\Delta \leftarrow \Delta + \mathbf{w} \cdot \boldsymbol{\ell}_t + \frac{1}{\eta} \ln(\mathbf{w} \cdot e^{-\eta \boldsymbol{\ell}_t})$ 
   $\mathbf{w} \leftarrow \mathbf{w} \odot e^{-\eta \boldsymbol{\ell}_t} / (\mathbf{w} \cdot e^{-\eta \boldsymbol{\ell}_t})$ 
end for

```

Lemma 5 (Paper's Lemma 4 — regret is governed by the number of segments). *If AdaHedge has started m segments after T rounds, then*

$$R_{\text{AdaHedge}}(T) < 2 \ln K \frac{\phi^m - 1}{\phi - 1} + m \left(\frac{\ln K}{e-1} + \frac{1}{8} \right).$$

Theorem 6 (Paper's Theorem 5 — worst case). $R_{\text{AdaHedge}}(T) \leq \frac{\phi \sqrt{\phi^2 - 1}}{\phi - 1} \sqrt{\frac{4}{e-1} L_T^* \ln K} + O((\ln L_T^* + 2) \ln K)$, with leading factor 3.46 at $\phi = 2$.

Throughout, the numbered results below are *this report's* numbering; each caption states the corresponding number in the paper. The easy-data half of the paper explains *when* m stays small, through the posterior mass $w_t^*(\eta) = \max_k w_t^k(\eta)$. Since $w_t^k \propto e^{-\eta L_{t-1}^k}$, this is the mass on the *current* leader $\arg \min_k L_{t-1}^k$, which need not be the hindsight best action.

Lemma 7 (Paper's Lemma 6). *For every t and $\eta \in (0, 1]$, $\delta_t(\eta) \leq (e - 2) \eta (1 - w_t^*(\eta))$.*

Theorem 8 (Paper's Theorem 9 — easy stochastic data). *If the loss vectors are independent and some action k^* satisfies $\min_{k \neq k^*} \mathbb{E}[\ell_t^k - \ell_t^{k^*}] \geq 2\alpha$ for all t , then with probability at least $1 - \delta$ AdaHedge starts a bounded number of segments and $R_{\text{AdaHedge}}(T) = O(K + \log(1/\delta))$.*

Note — *The chain the experiments are built to test.* Lemmas 7 and 5 chain into a single causal claim that is measurable end to end:

$$w_t^* \rightarrow 1 \text{ fast} \implies \Delta_T \text{ bounded} \implies \text{budget never depleted} \implies m \text{ small} \implies R(T) \text{ small.}$$

Section 7 measures every arrow separately rather than only the endpoints.

3 Implementation and correctness

Everything is written from scratch in a single Jupyter notebook using only `numpy` and `matplotlib`; no optimisation or machine-learning library is used. Two points deserve comment.

Hedge is vectorised, AdaHedge is not. Since $w_{t+1}^k \propto e^{-\eta L_t^k}$, the entire weight trajectory of fixed-rate Hedge is a row-wise softmax of the shifted cumulative-loss matrix and is produced in one pass, with the max subtracted for stability. AdaHedge cannot be vectorised: the segment boundaries depend on Δ , which depends on the weights, so it is an explicit T -round loop.

The mixability gap must be computed in centred form. The definition in (1) subtracts two nearly equal quantities. Writing $\bar{\ell}_t = \ell_t - \mathbf{w}_t \cdot \ell_t$, the η -dependent factor cancels and

$$\delta_t(\eta) = \frac{1}{\eta} \ln(1 + \mathbf{w}_t \cdot (e^{-\eta \bar{\ell}_t} - 1)), \quad (3)$$

evaluated with `log1p` and `expm1`. This matters because AdaHedge halves η at every restart and then *sums* δ_t over thousands of rounds, so a systematic sign error at small η would corrupt the restart decision itself. Form (3) also makes $\delta_t \geq 0$ structural rather than accidental: $\sum_k w_t^k \bar{\ell}_t^k = 0$, so Jensen gives $\mathbf{w}_t \cdot e^{-\eta \bar{\ell}_t} \geq 1$.

Three checks. On one stochastic instance:

- (a) the vectorised Hedge agrees with a naive round-by-round multiplicative-weights loop to $1.1 \cdot 10^{-15}$;
- (b) Lemma 3 holds every round: $\max_t \delta_t / (\eta/8)$ equals 1.0000, 0.9999, 0.9981, 0.9844 at $\eta = 0.005, 0.05, 0.25, 1$, never exceeding 1 (the smallest δ_t observed is $-3 \cdot 10^{-16}$, machine epsilon on a true zero);
- (c) the master bound (1) holds and is tight to two decimals at all four rates, e.g. $R(T) = 278.34$ against $\ln K / \eta + \Delta_T = 278.36$ at $\eta = 0.005$.

Check (b) is the strongest of the three: it is a per-round inequality, so a single violation in 4×10^4 rounds would falsify the implementation.

4 Experimental design

$T = 10^4$ rounds, $K = 4$ actions, losses in $\{0, 1\}$, 50 seeds fixed in advance. Every regime and every algorithm sees the *same* 50 loss matrices, so all comparisons are paired and the reported standard deviations measure only the variability of the data, not of the algorithms (Hedge and AdaHedge are both deterministic given a loss matrix).

Four algorithms are compared: AdaHedge with $\phi = 2$ and $\eta_1 = 1$, and Hedge at $2\eta^*, \eta^*, \eta^*/2$ where $\eta^* = \sqrt{2 \ln K / L_T^*}$ is read off each individual loss matrix. Hedge therefore uses post-hoc knowledge of L_T^* that AdaHedge does not have; the comparison is deliberately unfair in Hedge’s favour.

Six regimes are used: **stochastic** (one action Bern(0.3), three Bern(0.5)); **low-gap** (the same with 0.49 instead of 0.3); **adversarial**, the deterministic cycle in which action $t \bmod K$ takes loss 1 and the others 0; and three AP variants described in §5.

Note — *Why the weight plots are ranked, not indexed.* The identity of the best action varies from seed to seed, and in the AP regimes from window to window. Averaging w_t^k at a fixed index k would therefore average curves belonging to different roles and wash out exactly the structure of interest. Before averaging, actions are re-indexed inside each seed by their cumulative loss at $t = T$, so *rank 1* always means the best action in hindsight for that particular run.

5 The AP regime: a leader that switches on a known schedule

The two textbook regimes sit at the extremes: in **stochastic** one action is best throughout, in **adversarial** no action is ever better than another. AP (adversarial permutation) fills in the middle. It produces sequences

with a genuine leader inside each window, but an identity for that leader that changes at every window boundary, and a signal strength the designer controls.

5.1 Construction

Definition 9 (AP window). Fix a number of windows n , so each window has width $W = T/n$. Inside window i :

- (i) every action starts with exactly $W/2$ ones, placed as an independent uniform random permutation of its own column;
- (ii) a *leader* is drawn uniformly among the $K - 1$ actions that were not leader in window $i - 1$;
- (iii) x_i of the leader's ones are removed and redistributed as evenly as possible among the other $K - 1$ actions ($\lfloor x_i/(K - 1) \rfloor$ each, the remainder to randomly chosen distinct actions).

Three properties follow, and each is asserted at run time in the notebook.

Proposition 10 (Mass conservation). *Ones are moved, never created, so every window contains exactly $K \cdot W/2$ ones and the total loss mass is identical across the three variants and across seeds. Difficulty is therefore controlled purely by how the mass is arranged, not by how much of it there is.*

Proposition 11 (Deterministic endpoints, random paths). *Because each column carries an exact count, the cumulative losses at every window boundary are a deterministic function of the leader sequence: at the end of window i the leader lies $\frac{4}{3}x_i$ below each other action. Randomness affects the path inside a window, never the endpoint. This is what keeps the standard deviations in Table 3 small enough to compare regimes at 50 seeds.*

Proposition 12 (Turnover of the window leader). *Since the leader must differ from the previous window's leader, the window leader changes at every one of the $n - 1$ interior boundaries.*

Remark 13 (What that does not say). It does not say that the action with the smallest *cumulative* loss changes at those boundaries, and the two must not be conflated. Cumulative losses stay within tens of one another out of thousands, so the running $\arg \min_k L_t^k$ flickers whenever two actions are near-tied. Measured over 50 seeds it changes 74.2, 60.3 and 96.1 times in AP-10, AP-20 and AP-25, against only 9, 19 and 24 boundaries — an order of magnitude more often, and driven by near-ties rather than by the window schedule.

5.2 Why three variants: separating the level of x from its growth

The gap schedule is arithmetic, $x_i = x_1 + (i - 1)\Delta x$, and the three variants were chosen to occupy three different cells of the (initial level, growth rate) plane while ending at the *same* final relative gap of 20% of the window width. They are therefore three routes to the same destination, which is what makes them comparable.

Table 1: The three AP variants. All three end at $x_n/W = 20\%$; they differ in where they start and how fast they get there. The last column is the leader's cumulative advantage over each other action accumulated within a single window, $\frac{4}{3}x_i$.

	n	W	x_1	Δx	x_n	x_1/W	$\Delta x/W$	window gap $\frac{4}{3}x_i$
AP-10	10	1000	11	21	200	1.1%	2.10%	14.7 $\rightarrow$ 266.7
AP-20	20	500	43	3	100	8.6%	0.60%	57.3 $\rightarrow$ 133.3
AP-25	25	400	8	3	80	2.0%	0.75%	10.7 $\rightarrow$ 106.7

- **AP-10 — small x , large Δx .** The first window is almost pure noise (1.1% gap) and the last is unmistakable (20%). Few, long windows, and the difficulty collapses late and sharply.
- **AP-20 — large x , small Δx .** The signal is already strong in window 1 (8.6%) and barely strengthens. Twice as many leader switches, each of them immediately detectable.
- **AP-25 — both small.** A weak start and slow growth, spread over the largest number of the shortest windows: the leader changes 24 times, and for much of the horizon the current leader is only marginally identifiable.

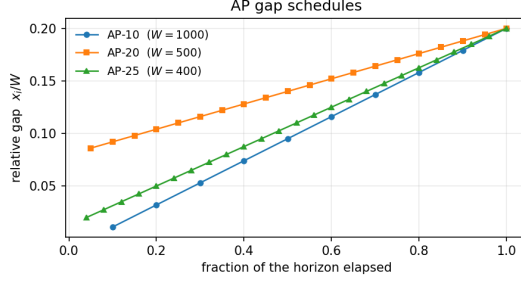


Figure 1: The three gap schedules on a common horizontal axis (fraction of the horizon elapsed). The three curves meet at 20% but approach it from different levels and slopes; AP-10 is the steep late-collapsing one, AP-20 the high flat one, AP-25 the low slow one.

5.3 AP results

The three variants order themselves exactly as the design intends (Table 3). AdaHedge attains 32.3 on AP-10, -6.5 on AP-20 and 48.0 on AP-25; the fixed rates are essentially flat across the three (44–52 at $2\eta^*$), so the variation is a property of AdaHedge, not of the data’s overall difficulty.

AP-20 is where AdaHedge does best and is the only regime with negative regret: on 27 of the 50 seeds it finishes *below the best single action*. That is not an error. The comparator in the regret is a fixed action held for all T rounds, whereas AP moves the leader every 500 rounds; an algorithm that shifts its weight in time can beat every fixed action, and the bounds of §2 are upper bounds that permit this. AP-20 is the variant in which the signal is strong from window 1, so the weight can move fast enough to exploit each switch — which is precisely the property the schedule was designed to isolate. AP-25 is the hardest of the three for the same reason read backwards: the smallest windows combined with the weakest early signal give the least time to react per switch.

Do the restarts track the leader changes?

The natural hypothesis is that AdaHedge restarts when the leader changes. It does not. Table 2 gives the mean round of the j -th segment start, and the mean distance from a restart to the nearest window boundary against the null value $W/4$ expected if restarts fell at rounds unrelated to the boundaries.

Table 2: AdaHedge restart timing in the AP regimes, over 50 seeds. Entries are mean $\pm$ sd of the round at which segment j begins; a parenthesised count marks a segment reached on only some seeds. The observed distance to the nearest boundary is *larger* than the null in all three variants.

	$j = 2$	$j = 3$	$j = 4$	$j = 5$	$j = 6$
AP-10 ($W=1000$)	78 ± 214	244 ± 259	678 ± 315	3021 ± 1290	—
AP-20 ($W=500$)	275 ± 386	919 ± 326	2697 ± 1191 (49)	6266 ± 1533 (37)	—
AP-25 ($W=400$)	67 ± 81	219 ± 134	638 ± 207	2075 ± 324	8793 ± 727 (16)
mean distance to nearest boundary / null $W/4$:	AP-10 588/250	AP-20 179/125	AP-25 188/100		

The explanation is quantitative and comes from Lemma 3. Since $\delta_t \leq \eta/8$, exhausting the budget (2) takes at least

$$T_{\min}(\eta) = \frac{b(\eta)}{\eta/8} = \frac{8 \ln K}{\eta^2} + \frac{8 \ln K}{(e-1)\eta} \quad (4)$$

rounds *whatever the data are*. With $K = 4$ and $\eta_i = 2^{1-i}$ this gives $T_{\min} = 17.5, 57.3, 203.3, 761.4, 2942.4$ for the first five segments, so the earliest possible restart rounds are 17.5, 74.8, 278.1, 1039.5, 3981.9 — a *geometric* ladder fixed by ϕ alone. The observed means (78, 244, 678, 3021 for AP-10) sit just above that ladder. The window boundaries, by contrast, form an *arithmetic* ladder (1000, 2000, $\dots$). Two ladders of different type cannot align, and the restarts follow the one belonging to the algorithm, not the one belonging to the data.

Note — *Reading this as a limitation, not a failure.* AdaHedge is not a change-point detector and does not claim to be. What (4) shows is that the doubling trick makes it *quadratically* more reluctant to restart as η falls, so by construction it cannot respond to a switching leader late in the horizon. This is the same structural cost that the 2014 successor [2] removes by letting $\eta_t = \ln K / \Delta_{t-1}$ vary continuously, with no resets at all.

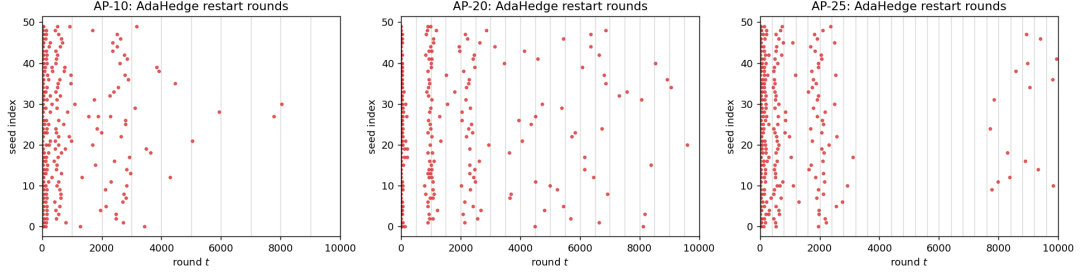


Figure 2: Every AdaHedge restart, one row per seed; grey vertical lines are the window boundaries at which the leader changes. Restarts crowd into the first few hundred rounds and then thin out geometrically, independently of where the boundaries fall.

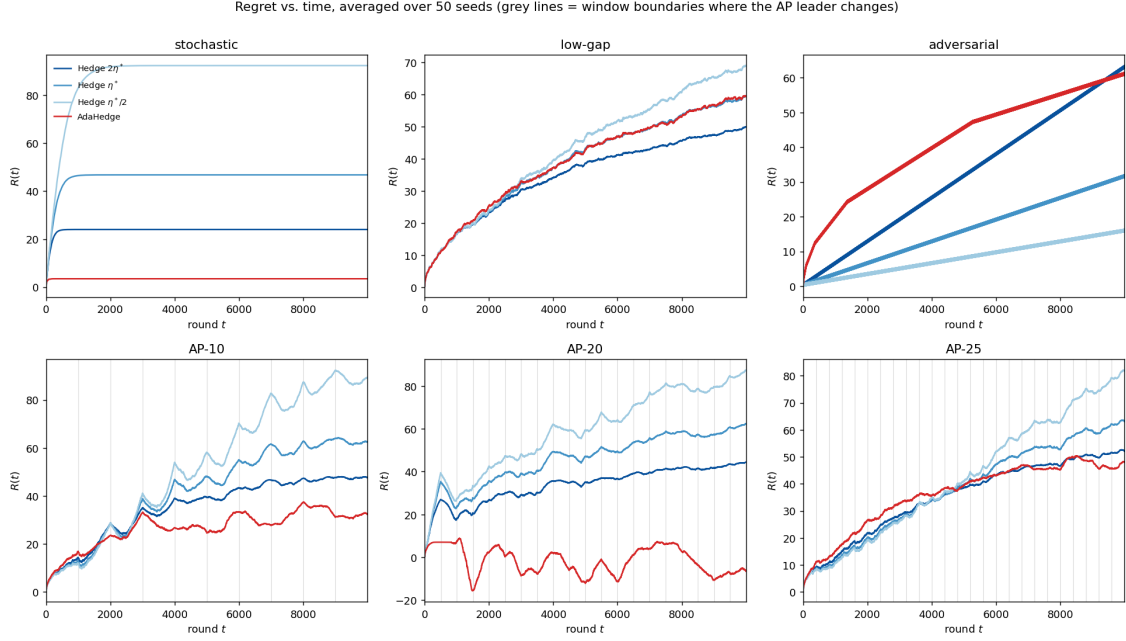


Figure 3: Regret against time, averaged over 50 seeds, with the comparator $\min_k L_t^k$ recomputed at every t . Grey verticals mark AP window boundaries. Note the sign reversal of the η -ordering between the top-left and top-right panels, the flat AdaHedge curve in **stochastic**, the kinks in the AdaHedge curve where it restarts, and the excursion below zero in **AP-20**.

6 Results

Table 3: Final regret $R(T)$, mean $\pm$ sd over 50 seeds. m is the mean number of AdaHedge segments and η_{fin} its mean final learning rate. Best entry per row in bold. The **adversarial** regime is deterministic, hence zero standard deviation.

regime	Hedge $2\eta^*$	Hedge η^*	Hedge $\eta^*/2$	AdaHedge	L_T^*	η^*	m	η_{fin}
stochastic	24.0 ± 0.3	46.8 ± 0.4	92.4 ± 0.6	3.5 ± 2.3	2997	0.0304	1.18	0.915
low-gap	50.0 ± 6.0	59.6 ± 8.4	69.0 ± 18.4	59.7 ± 22.8	4896	0.0238	5.46	0.055
adversarial	62.8 ± 0.0	31.3 ± 0.0	15.6 ± 0.0	60.8 ± 0.0	2500	0.0333	6.00	0.031
AP-10	47.6 ± 4.9	62.4 ± 10.9	89.2 ± 23.9	32.3 ± 22.7	4817	0.0240	5.00	0.063
AP-20	44.4 ± 5.5	62.3 ± 8.4	87.4 ± 19.5	-6.5 ± 44.6	4845	0.0239	4.72	0.081
AP-25	52.1 ± 5.7	63.1 ± 10.8	81.8 ± 23.0	48.0 ± 28.8	4871	0.0239	5.32	0.053

7 Reading the results against the theory

7.1 The chain, link by link

Table 4 measures every arrow of the chain stated in §2, using $\eta = 1$ throughout so that the budget $b(1) = 2.193$ is a single fixed threshold.

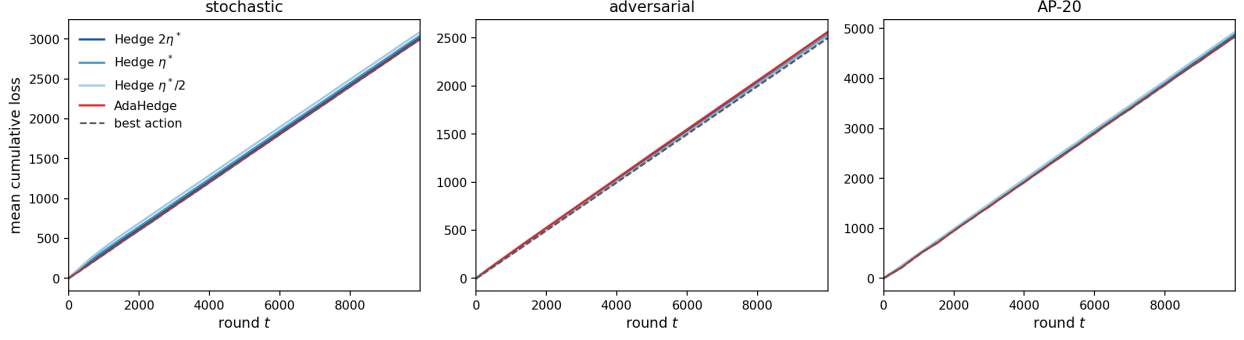


Figure 4: Cumulative loss against time, averaged over 50 seeds, with the running best action dashed, for three representative regimes. All four curves are visually indistinguishable: the total losses are thousands and they differ by tens. That is the whole reason regret, and not loss, is the quantity of interest — Figure 3 plots the same information after the comparator has been subtracted. The notebook produces this figure for all six regimes, together with the difference of these curves.

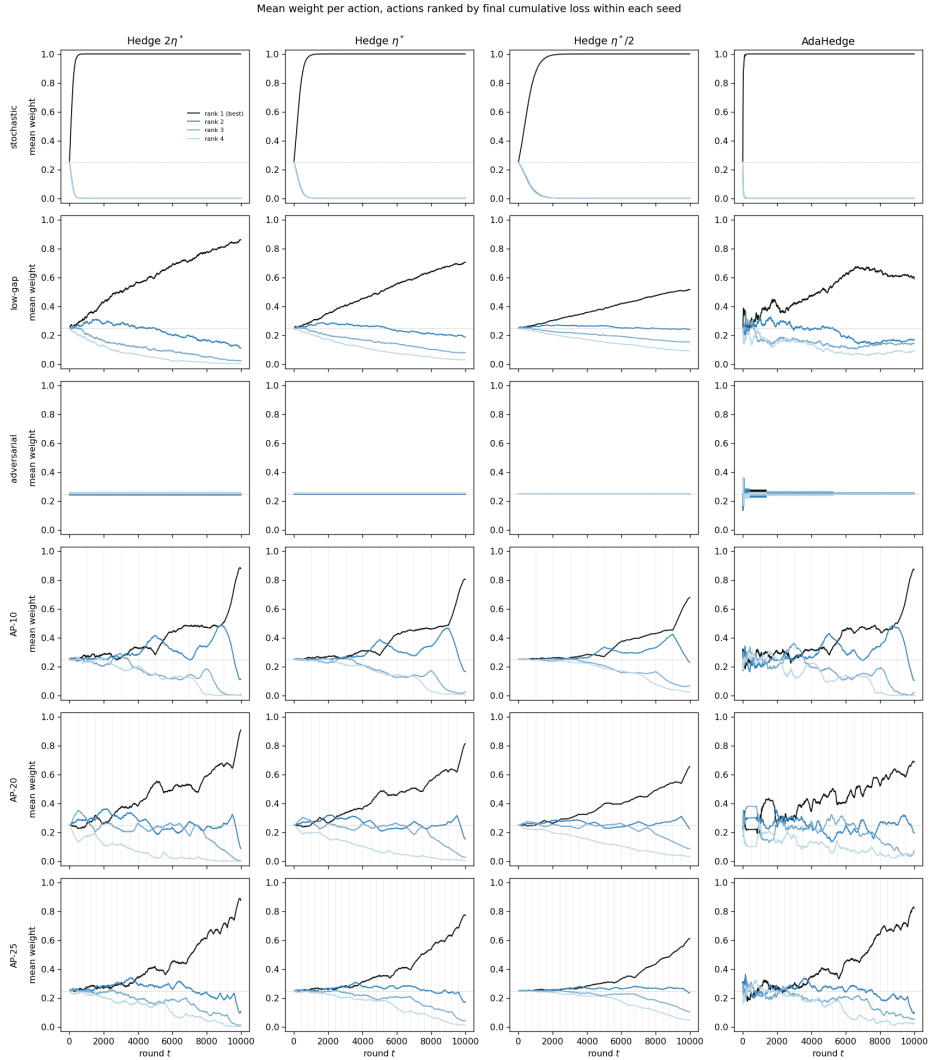


Figure 5: Mean weight placed on each action against time, with actions ranked inside each seed by their final cumulative loss (rank 1 = best in hindsight). Rows are regimes, columns are algorithms. The dotted line is $1/K$. Read the rows: **stochastic** converges to a point mass within a few hundred rounds under every algorithm; **adversarial** never leaves $1/K$; the AP rows show the rank-1 action separating only in the second half of the horizon, as the gap schedule intends.

Table 4: Every link of the chain $w^* \rightarrow \Delta_T \rightarrow \text{budget} \rightarrow m \rightarrow R$, measured. w_T^* is the mean weight on the *hindsight-best* action at $t = T$ under Hedge($2\eta^*$) — sense (i) of Note 1. It agrees with $\max_k w_T^k$ in every regime but **adversarial**, where the four actions tie and the two read 0.246 and 0.263, both $\approx 1/K$; $\Delta_T(1)$ is the cumulative mixability gap of Hedge run at $\eta = 1$, to be compared with $b(1) = 2.193$ and with the Lemma 3 ceiling $\eta T/8 = 1250$.

regime	w_T^*	$\Delta_T(1)$	budget $b(1) = 2.193$	m	R_{AdaHedge}
stochastic	1.000	1.54	never depleted	1.18	3.5
AP-20	0.910	29.16	depleted	4.72	−6.5
AP-10	0.878	36.30	depleted	5.00	32.3
low-gap	0.863	43.22	depleted	5.46	59.7
AP-25	0.874	49.28	depleted	5.32	48.0
adversarial	0.246	969.55	depleted	6.00	60.8

Reading top to bottom, the posterior mass on the best action falls from 1.000 to 0.246, the cumulative mixability gap rises from 1.54 to 969.55 — a factor of 630 — the number of segments rises from 1.18 to 6, and the regret rises from 3.5 to 60.8. Lemma 7 predicts exactly this monotone relation between the first two columns, and Lemma 5 the relation between the last two. Two entries deserve separate comment.

- In **stochastic**, $\Delta_T(1) = 1.54$ against a budget of 2.193: the budget is *never* depleted, so AdaHedge stays on one segment at $\eta = 1$ for the whole horizon. This is Theorem 8 observed directly rather than inferred from the regret — and it is confirmed by the segment histogram, $m = 1$ on 42 of 50 seeds, $m = 2$ on seven, $m = 3$ on one.
- In **adversarial**, $\Delta_T(1) = 969.55$ against the Lemma 3 ceiling $\eta T/8 = 1250$: the data realise 78% of the worst gap that any sequence can produce. The corresponding $w_T^* = 0.246 \approx 1/K$ says the posterior never converges at all, which is the paper’s own characterisation of a hard instance.

Remark 14 (The worst-case ceilings are the wrong tool on easy data, by design). On **stochastic** the two worst-case ceilings give $\Delta_T(1) \leq 1250$ (Lemma 3) and $\Delta_T(1) \leq 1745$ (Lemma 4), against a measured 1.54. Both are loose by roughly three orders of magnitude. That is not a defect: those lemmas exist to make the *worst-case* guarantee go through, and Lemma 7 together with Theorem 8 exists precisely because they say nothing useful here.

7.2 Are the guarantees satisfied?

Lemma 5 and Theorem 6 were checked on each of the $6 \times 50 = 300$ (regime, seed) pairs individually, not only on the averages. Both hold on all 300. The Lemma 5 bound takes the values 3.7, 10.2, 22.2, 45.3, 90.6, 180.3 at $m = 1, \dots, 6$; the measured regrets are 3.5 at $m = 1.18$ (**stochastic**) — within 6% of the bound, essentially tight — and 60.8 at $m = 6$ (**adversarial**), a factor 3 inside it. Theorem 6’s leading term is 311–436 across the regimes, comfortably above every observed value, as a worst-case bound should be.

7.3 The reversal, and what it says about fixed rates

The single most important comparison in the table is between the first and third rows. In **stochastic** the regret *decreases* in η : $92.4 \rightarrow 46.8 \rightarrow 24.0$. In **adversarial** it *increases*: $15.6 \rightarrow 31.3 \rightarrow 62.8$. The ordering of the same three algorithms reverses completely. No fixed learning rate can be right in both regimes, and a practitioner tuning η on one kind of data would choose the worst of the three for the other. This is the entire experimental content of the paper, and it reproduces cleanly.

The mechanism in **adversarial** is worth stating because it is counter-intuitive. Hedge lowers the weight of whichever action was just hit, and in a cycle the action about to be hit is precisely the one that has been spared longest and therefore carries the highest weight. Larger η means a sharper — and systematically wrong — tilt, so the regret grows linearly in η . The weights oscillate with period K around $1/K$ with amplitude of order η/K , which at $\eta \approx 0.03$ is a few thousandths and invisible in Figure 5; only AdaHedge’s opening segments, run at η near 1, show visible oscillation.

7.4 Where AdaHedge loses

AdaHedge is not uniformly better, and the two regimes where it loses have different causes.

On **adversarial** it reaches 60.8 against 15.6 for the best fixed rate, a factor of 3.9. Lemma 5 names the cause: $m = 6$ segments on every seed, and each restart discards the posterior and re-pays $\ln K$ at a rate that has been halved again. The $2 \ln K(\phi^m - 1)/(\phi - 1)$ term is geometric in m , so the sixth restart alone costs more than the first five together. The cost is intrinsic to the doubling trick, not to the budget rule.

On **low-gap** it reaches 59.7 against 50.0, but here the limitation is statistical rather than algorithmic. The signal is $0.01 \cdot T = 100$ and the noise is of order $\sqrt{T} = 100$: the best action is barely identifiable in

principle, every algorithm sits between 50 and 69, and the standard deviations (6.0 to 22.8) are large relative to the differences between algorithms. The honest reading is that this regime does not discriminate.

8 Conclusions

The claims of the paper reproduce, and reproduce for the stated reasons rather than by coincidence. On easy data AdaHedge attains $R(T) = 3.5$, essentially the constant regret of Theorem 8, and does so by never depleting its budget — $\Delta_T(1) = 1.54$ against $b(1) = 2.193$ — which is the mechanism of Lemma 7 and Theorem 8, observed rather than inferred. On antagonistic data it remains within its worst-case guarantee while the fixed rates reverse their ordering entirely. Every per-round and per-run bound tested (Lemma 3, the master bound, Lemma 5, Theorem 6) holds without exception across 300 runs.

Three findings go beyond a confirmation.

1. **AdaHedge does not dominate.** It is $3.9\times$ the best fixed rate on the cyclic adversary, and Lemma 5 identifies the six restarts as the cause. Adaptivity is bought, not free, and the price is geometric in the number of restarts.
2. **The restarts are governed by the algorithm, not by the data.** Equation (4) forces the restart times onto a geometric ladder 17.5, 74.8, 278, 1040, 3982 set by ϕ alone; the AP window boundaries form an arithmetic ladder. The observed restarts follow the former, which is why AdaHedge cannot track a leader that switches late in the horizon.
3. **Negative regret is attainable and informative.** On AP-20, AdaHedge finishes below the best single action on 27 of 50 seeds. The comparator is fixed while the data’s leader moves, and AP-20 — strong signal from the first window, small subsequent growth — is the variant in which the weights can move fast enough to exploit each switch. AP-25, with the weakest early signal and the shortest windows, is the hardest of the three for exactly the mirrored reason.

The natural continuation is the 2014 successor [2], which replaces the doubling trick with a continuously varying $\eta_t = \ln K / \Delta_{t-1}$: the quantity that is merely *monitored* here goes on to *define* the learning rate, and no reset is needed — which would remove precisely the cost identified in findings 1 and 2 above.

Reproducibility

All results come from a single notebook, `adahedge_experiments.ipynb`, using only `numpy` and `matplotlib`. The 50 seeds are listed explicitly in the first code cell; every regime and algorithm consumes the same seed list, and the full run takes about 35 seconds and is bitwise reproducible. Numerical results are also written to `results.json`. Code and report: https://github.com/Ali-Hassanzadeh1995/ML-Project_2025-26.

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